DC 400W 15A Step-up Boost Converter

Model: 400W Boost Constant Current Module

Brand: HZWDONE

1. Product Overview

The HZWDONE 400W Boost Constant Current Module is a non-isolated step-up (BOOST) converter designed to efficiently increase a DC input voltage to a higher, adjustable DC output voltage. It features both voltage and current regulation, making it suitable for a wide range of applications including DIY power supplies, charging various battery types, and driving high-power LEDs.



Figure 1: HZWDONE 400W Boost Constant Current Module

2. Specifications

Feature	Specification
Module Name	400W Boost Constant Current Module
Module Properties	Non-isolated step-up module (BOOST)
Input Voltage	DC 8.5V - 50V
Input Current	15A (MAX), enhance heat dissipation if exceeding 8A

Feature	Specification
Quiescent Current	10mA (at 12V input, 20V output; increases with higher output voltage)
Output Voltage	10V - 60V (continuously adjustable)
Constant Current Range	0.2A - 12A
Operating Temperature	-40°C to +85°C (enhance heat dissipation if ambient temperature is too high)
Operating Frequency	150KHz
Conversion Efficiency	Up to 96%
Overcurrent Protection	Yes
Input Reverse Polarity Protection	No
Installation	4 x 2.55mm screw holes
Connection Method	Screw terminals for Input and Output
Module Dimensions (LxWxH)	67mm x 48mm x 28mm
Module Weight	90g
Package Dimensions (LxWxH)	7cm x 6cm x 4cm
Package Weight	0.067kg

3. Setup and Connection

Before connecting the module, ensure your power source and load are within the specified voltage and current ranges. This module does NOT have input reverse polarity protection, so careful attention to polarity is crucial.

Figure 2: Module Layout with Input/Output and Adjustment Points

1. **Identify Terminals:** Locate the input terminals (IN+, IN-) and output terminals (OUT+, OUT-) on the module. Refer to Figure 2.
2. **Connect Input Power:** Connect your DC power source to the IN+ and IN- terminals. Ensure correct polarity (IN+ to positive, IN- to negative). Incorrect connection will damage the module.
3. **Connect Load:** Connect your load to the OUT+ and OUT- terminals.
4. **Initial Adjustment (Voltage):** Before applying full power to your load, it is recommended to pre-adjust the output voltage.
    - Connect a voltmeter to the OUT+ and OUT- terminals.
    - Apply input power (e.g., 12V).
    - Turn the **CV-ADJ** potentiometer (Voltage Adjustment) clockwise to increase the output voltage, and counter-clockwise to decrease it. Adjust to your desired output voltage.
5. **Initial Adjustment (Current - if needed):** If you require constant current operation (e.g., for LED driving or battery charging), you will need to adjust the current limit.
    - Connect an ammeter in series with the output, or use a suitable load that allows current measurement.
    - Turn the **CC-ADJ** potentiometer (Current Adjustment) counter-clockwise to set the minimum current.
    - Then, turn it clockwise to increase the current to your desired limit.
    - **Note:** The constant current function works when the load current reaches the set current limit. If the load current is below the set limit, the module operates in constant voltage mode.
6. **Heat Dissipation:** If the input current exceeds 8A or if the module operates at high power for extended periods, additional heat dissipation (e.g., a fan) may be required to prevent overheating.

4. Operating Instructions

Once the module is set up and adjusted, it will maintain the set output voltage and current limit. Monitor the module's temperature during operation, especially under heavy loads.

4.1. Typical Applications

- **DIY Adjustable Power Supply:** Convert a fixed DC input (e.g., 12V) into an adjustable output (10-60V) for various electronic projects.
- **Electronic Device Power Supply:** Provide a stable and regulated power source for specific electronic devices requiring a higher voltage than available.
- **Vehicle Power Supply:** Use as a car power supply for laptops, PDAs, or other digital products.
- **Mobile Notebook Power Supply:** Combine with a high-capacity 12V lithium battery pack to create a portable power solution.
- **Solar Panel Regulator:** Regulate the output from solar panels to charge batteries or power devices.
- **Battery Charging:** Charge various types of batteries (e.g., lithium batteries) by setting appropriate voltage and current limits.
- **High-Power LED Driver:** Drive high-power LED lights with constant current regulation.

5. Maintenance

The 400W Boost Constant Current Module is designed for reliable operation with minimal maintenance. Follow these guidelines to ensure longevity:

- **Keep Clean:** Ensure the module is free from dust, dirt, and moisture. Use a soft, dry brush or compressed air to clean if necessary.
- **Environmental Conditions:** Operate the module within its specified temperature range (-40°C to +85°C). Avoid extreme humidity or corrosive environments.
- **Secure Connections:** Periodically check that all wire connections to the screw terminals are secure to prevent intermittent operation or arcing.
- **Ventilation:** Ensure adequate airflow around the module, especially if operating at higher currents or in enclosed spaces, to facilitate heat dissipation.

6. Troubleshooting

If you encounter issues with your module, refer to the following troubleshooting steps:

Problem	Possible Cause	Solution
No output voltage	• No input power • Incorrect input polarity • Faulty connections • Output voltage set too low	• Verify input power supply is active. • Check input wiring for correct polarity (IN+ to positive, IN- to negative). • Ensure all wires are securely connected to terminals. • Adjust CV-ADJ potentiometer clockwise to increase output voltage.
Output voltage is incorrect or unstable	• CV-ADJ potentiometer not correctly set • Input voltage fluctuations • Overload condition	• Re-adjust CV-ADJ potentiometer with a voltmeter. • Ensure input power supply is stable. • Reduce load or ensure load current is within specifications.
Module overheating	• Excessive input current (above 8A) • Insufficient heat dissipation • High ambient temperature	• Reduce input current or load. • Add external cooling (e.g., a fan) to the heatsinks. • Ensure module is in a well-ventilated area.
Output current is not limited as expected	• CC-ADJ potentiometer not correctly set • Load current is below set limit	• Adjust CC-ADJ potentiometer with an ammeter in series with the load. • The constant current function only activates when the load attempts to draw more current than the set limit.

7. User Tips

- **Pre-adjustment:** Always adjust the output voltage and current limit (if applicable) with no load or a dummy load before connecting your sensitive device.
- **Polarity Check:** Double-check input polarity before powering on, as the module lacks reverse polarity protection.
- **Off-Grid Use:** This module is well-suited for off-grid applications, such as boosting voltage from solar panels or 12V battery packs for various devices.
- **Current Monitoring:** When operating near the maximum current (15A input, 12A output), closely monitor the module's temperature and ensure adequate cooling.

8. Frequently Asked Questions (FAQ)

Q: Can this module be used in places without a power grid, like in a field or at sea?

A: Yes, this module is suitable for such applications. It can boost voltage from a 12V battery pack or solar panel to power various electronic devices, making it ideal for portable or off-grid setups.

Q: What is the maximum output current?

A: The maximum constant current range is 12A. The maximum input current is 15A, but if the input current exceeds 8A, enhanced heat dissipation is required.

Q: Does this module have reverse polarity protection?

A: No, the module does not have input reverse polarity protection. Ensure correct input wiring to avoid damage.

9. Warranty and Support

For warranty information and technical support, please contact the seller or manufacturer directly. Keep your purchase records for reference.